

Load Interconnection System Impact Study Report

For Queue Project Q22-2081

Clover 345 kV

July 03, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2081. The Interconnection Customer proposes a 451.8 MW large load (data center) facility interconnecting at the Clover 345 kV substation in the Virginia area. The request was received on May 11, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 673 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$461,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Barr Lake to Kean Mtn And Grassy Crk 345.0 kV L	Raland to North Pt 345.0 kV	102.3	1,240	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Barr Lake to Kean Mtn And Grassy Crk 345.0 kV L	Reconductor line	\$17,896,000	3% (\$461,000)
	Total allocated to project		\$461,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.